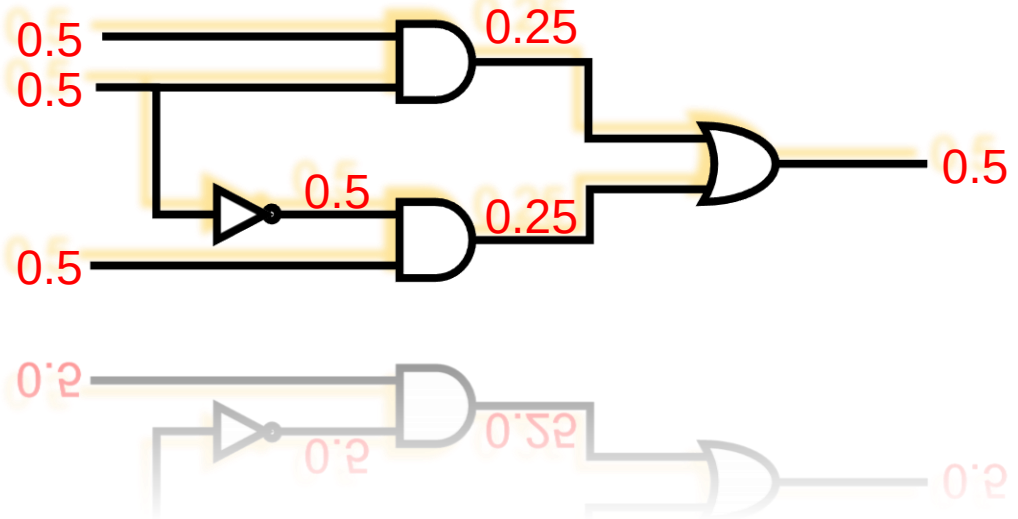


Testability Measure

- Introduction
- **SCOAP**
 - ♦ Combinational
 - ♦ **Sequential**
- COP
- High-level testability measures



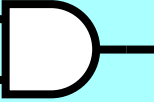
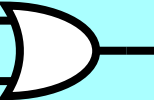
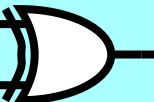
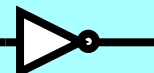
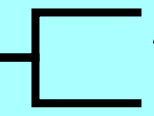
Sequential SCOAP Measures

- **Sequential controllability: $SC^0(N)$, $SC^1(N)$**
 - ♦ Minimum number of **FF assignments** (number of clock cycles) required to control 0 or 1 on node N
 - ♦ **smaller** number means **easier** to control
- **Sequential observability: $SO(N)$**
 - ♦ Minimum number of **FF assignments** required to propagate logical value on node N to a primary output
- **NOTE: assume no scan**
 - ♦ Can only control PI, observe PO
 - ♦ Can **NOT** control FF, can **NOT** observe FF

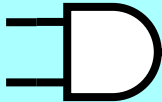
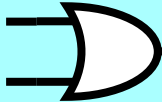
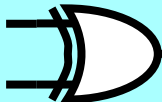


**Sequential SCOAP Measures
of Clock Cycles Needed**

SC⁰(N) and SC¹(N)

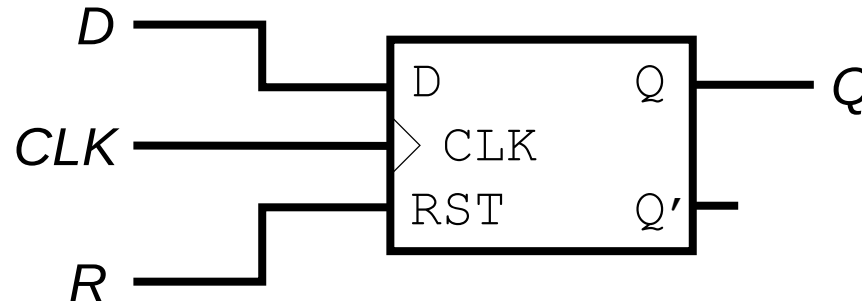
	SC ⁰ (y)	SC ¹ (y)
Primary inputs	0 (not 1)	0
X_1 X_2  y	$\min[SC^0(x_1), SC^0(x_2)]$ +1	$SC^1(x_1) + SC^1(x_2)$
X_1 X_2  y	$SC^0(x_1) + SC^0(x_2)$	$\min[SC^1(x_1), SC^1(x_2)]$
X_1 X_2  y	$\min[SC^0(x_1) + SC^0(x_2),$ $SC^1(x_1) + SC^1(x_2)]$	$\min[SC^0(x_1) + SC^1(x_2),$ $SC^1(x_1) + SC^0(x_2)]$
x  y	$SC^1(x)$	$SC^0(x)$
X_1  y_1 y_2	$SC^0(y_1) = SC^0(y_2) = SC^0(x_1)$	$SC^1(y_1) = SC^1(y_2) = SC^1(x_1)$

SO(N)

	SO(x_1)
Primary outputs	0
x_1 x_2  y	SO(y) + SC ¹ (x_2) +1
x_1 x_2  y	SO(y) + SC ⁰ (x_2)
x_1 x_2  y	SO(y) + min[SC ⁰ (x_2), SC ¹ (x_2)]
x_1  y	SO(y)
x_1  y_1 y_2	min[SO(y_1), SO(y_2)]

Flip-Flop (Controllability)

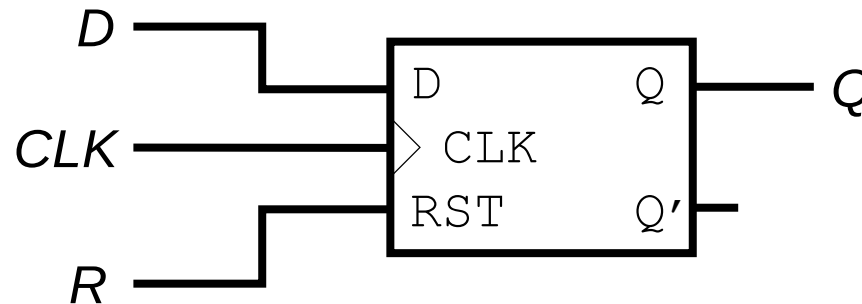
- Positive edge triggered, asynchronous reset



$$CC^1(Q) = CC^1(D) + CC^1(CLK) + CC^0(CLK) + CC^0(R)$$
$$SC^1(Q) = SC^1(D) + SC^1(CLK) + SC^0(CLK) + SC^0(R) + 1$$

$$CC^0(Q) = \min[CC^1(R),$$
$$CC^0(D) + CC^1(CLK) + CC^0(CLK) + CC^0(R)]$$
$$SC^0(Q) = \min[SC^1(R),$$
$$SC^0(D) + SC^1(CLK) + SC^0(CLK) + SC^0(R)] + 1$$

Flip-Flop (Observability)



$$\begin{aligned} CO(D) &= CO(Q) + CC^1(CLK) + CC^0(CLK) + CC^0(R) \\ SO(D) &= SO(Q) + SC^1(CLK) + SC^0(CLK) + SC^0(R) + 1 \end{aligned}$$

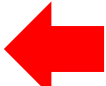
Seq. SCOAP Computation Alg.

- Computation of SC, SO is similar to CC, CO
 - ♦ but require *iterations* for controllability to converge

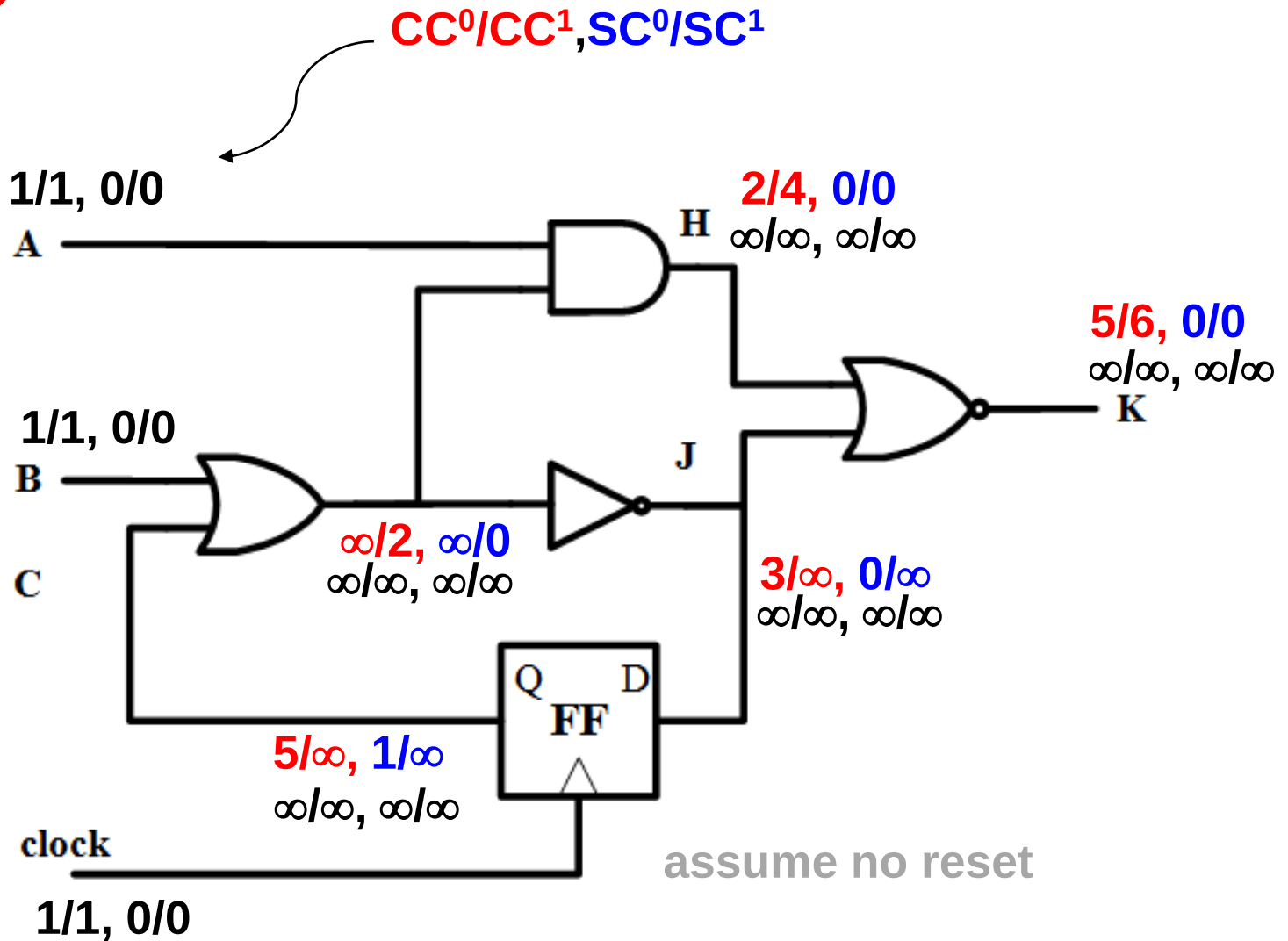
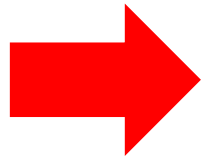
Controllability:

1. For all PI's, set $CC^0 = CC^1 = 1$ and $SC^0 = SC^1 = 0$
2. For all other nodes, set $CC^0 = CC^1 = \infty$ and $SC^0 = SC^1 = \infty$
3. Propagate controllability from PI's to PO's
Iterate until numbers stabilize. 

Observability:

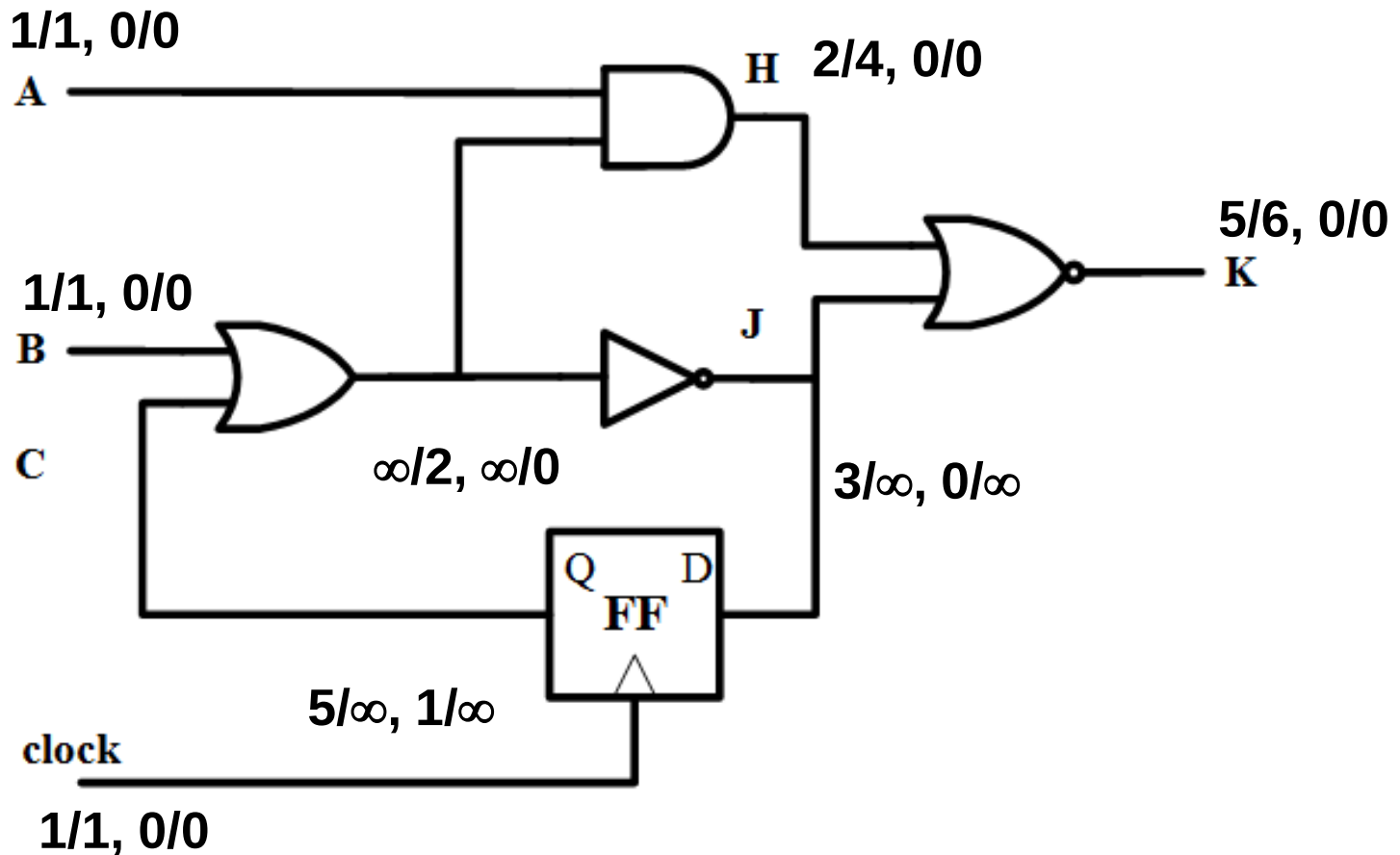
1. For all PO's, set $CO = SO = 0$
2. For all other nodes, set $CO = SO = \infty$
3. Propagate observability from PO's to PI's
(note: no iteration needed for CO/SO) 

Controllability Computation - 1

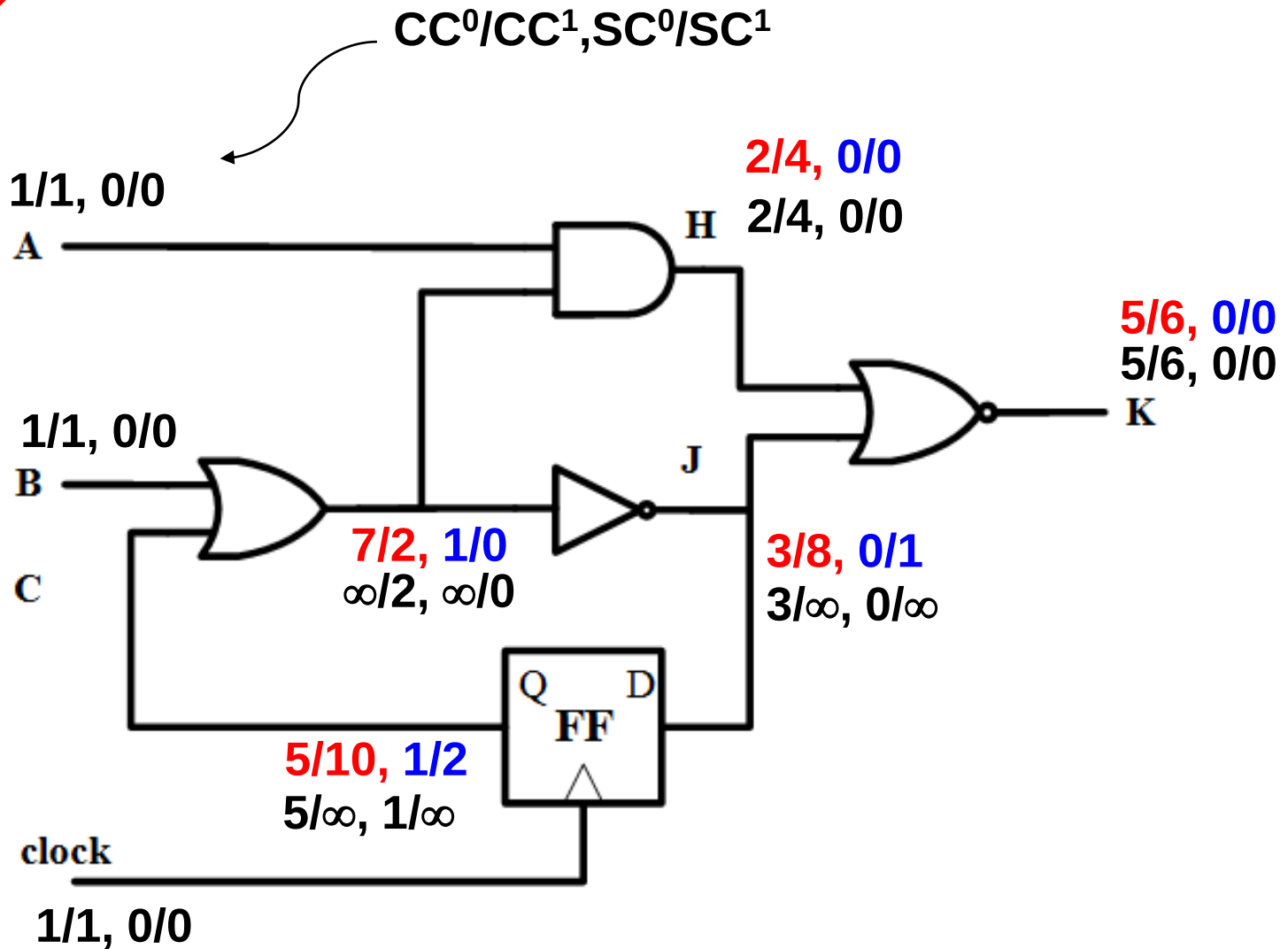
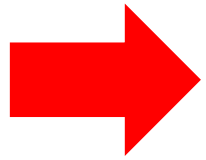


Quiz

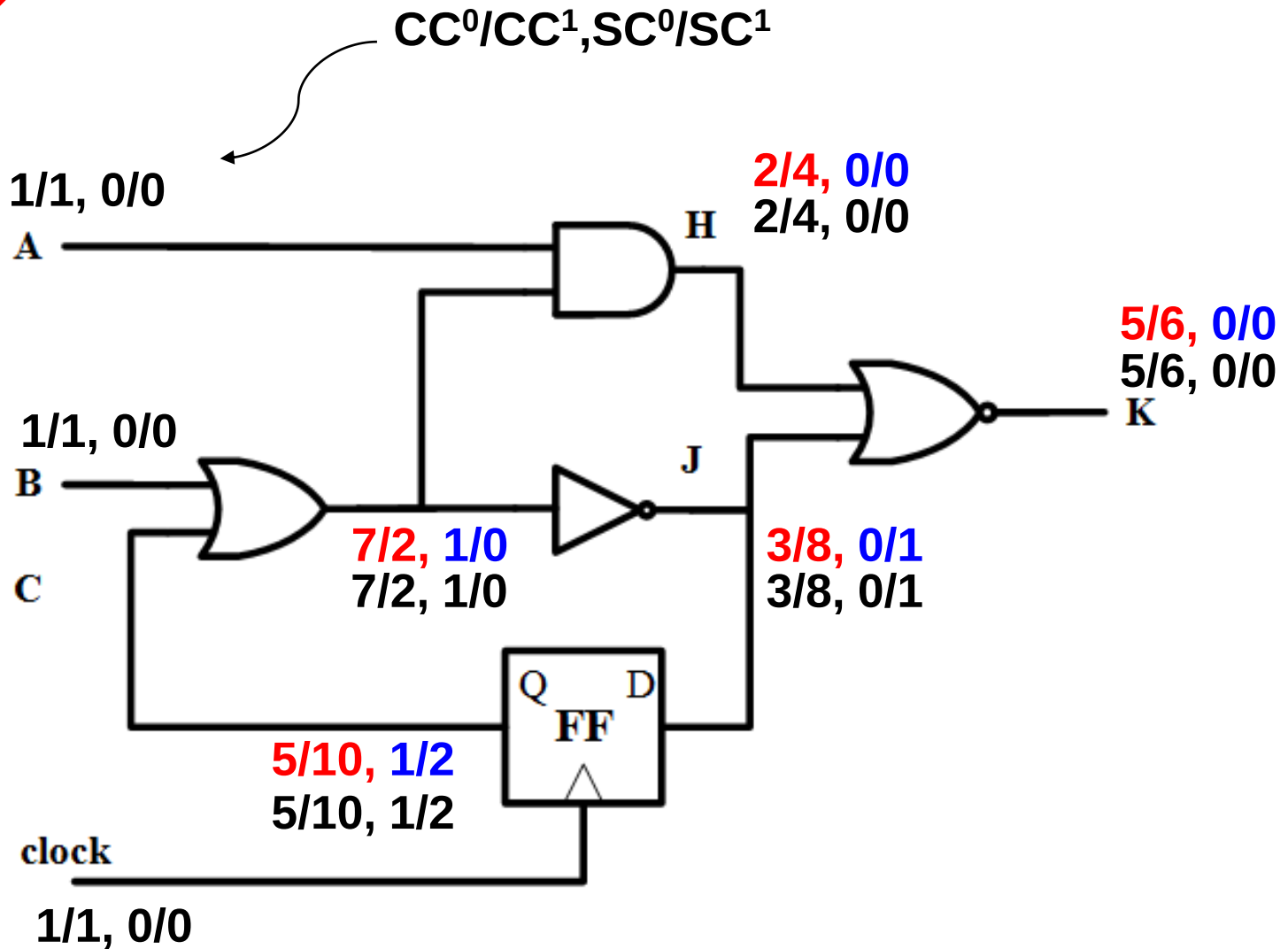
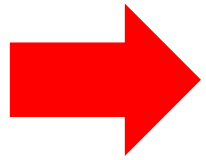
Q: Given numbers from the 1st iteration, please continue to calculate CC^0/CC^1 , SC^0/SC^1 in 2nd iteration.



Controllability Computation - 2

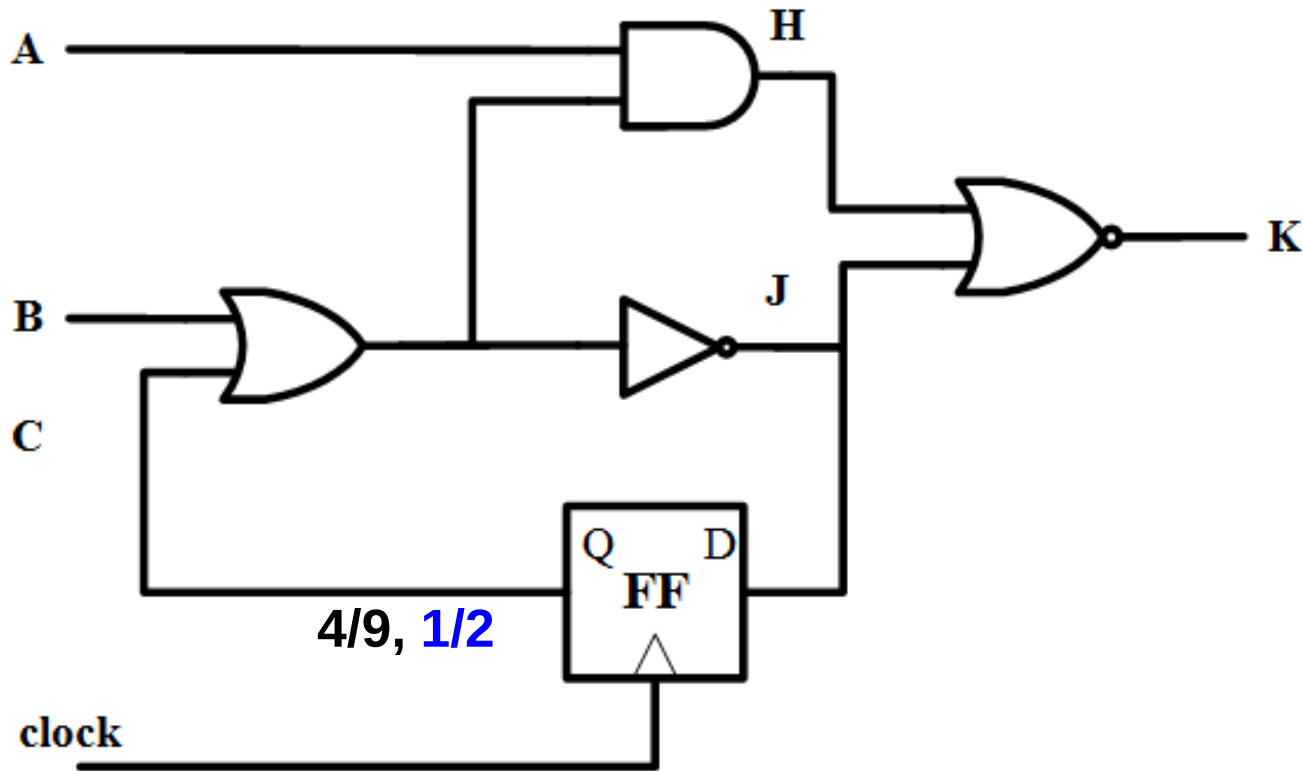


Controllability Computation - 3



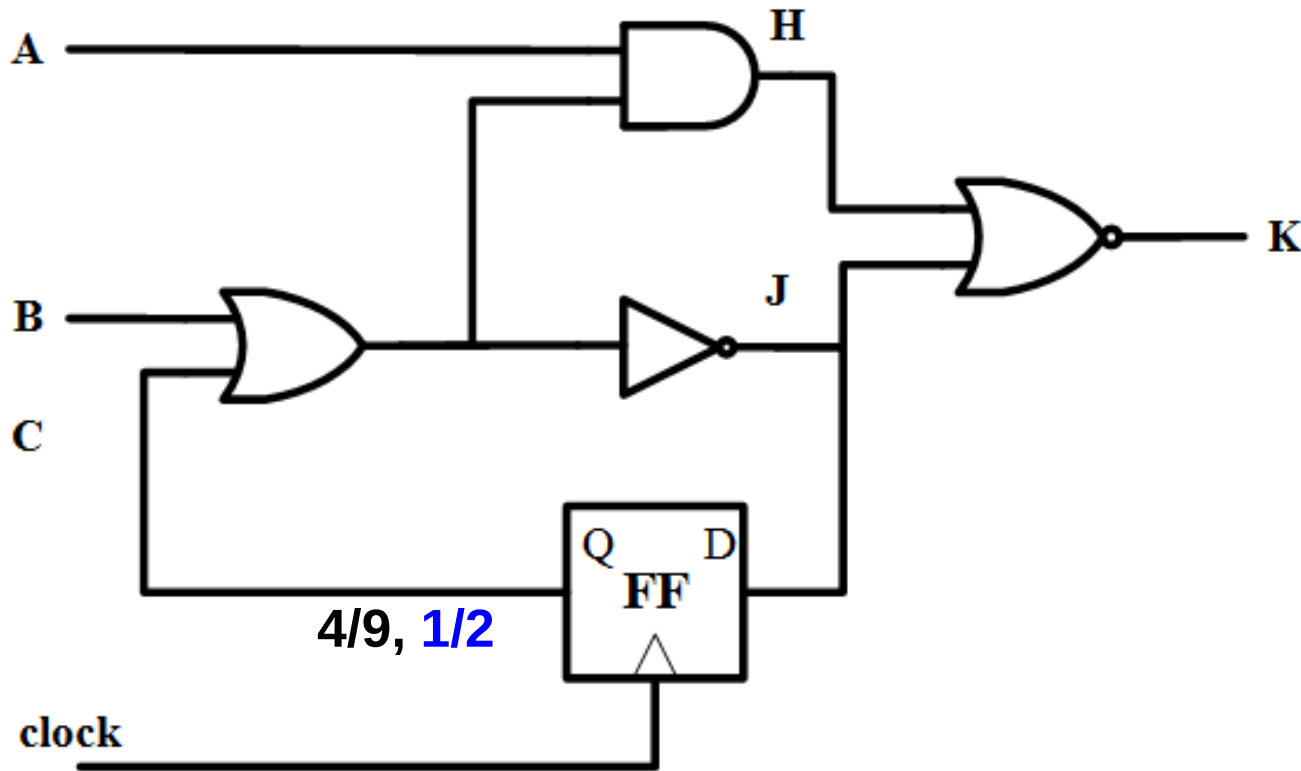
Quiz

Q1: Generate a sequence of test patterns to control C to 0?
Q2: Generate a sequence of test patterns to control C to 1?
(assume no scan. can only assign PI)

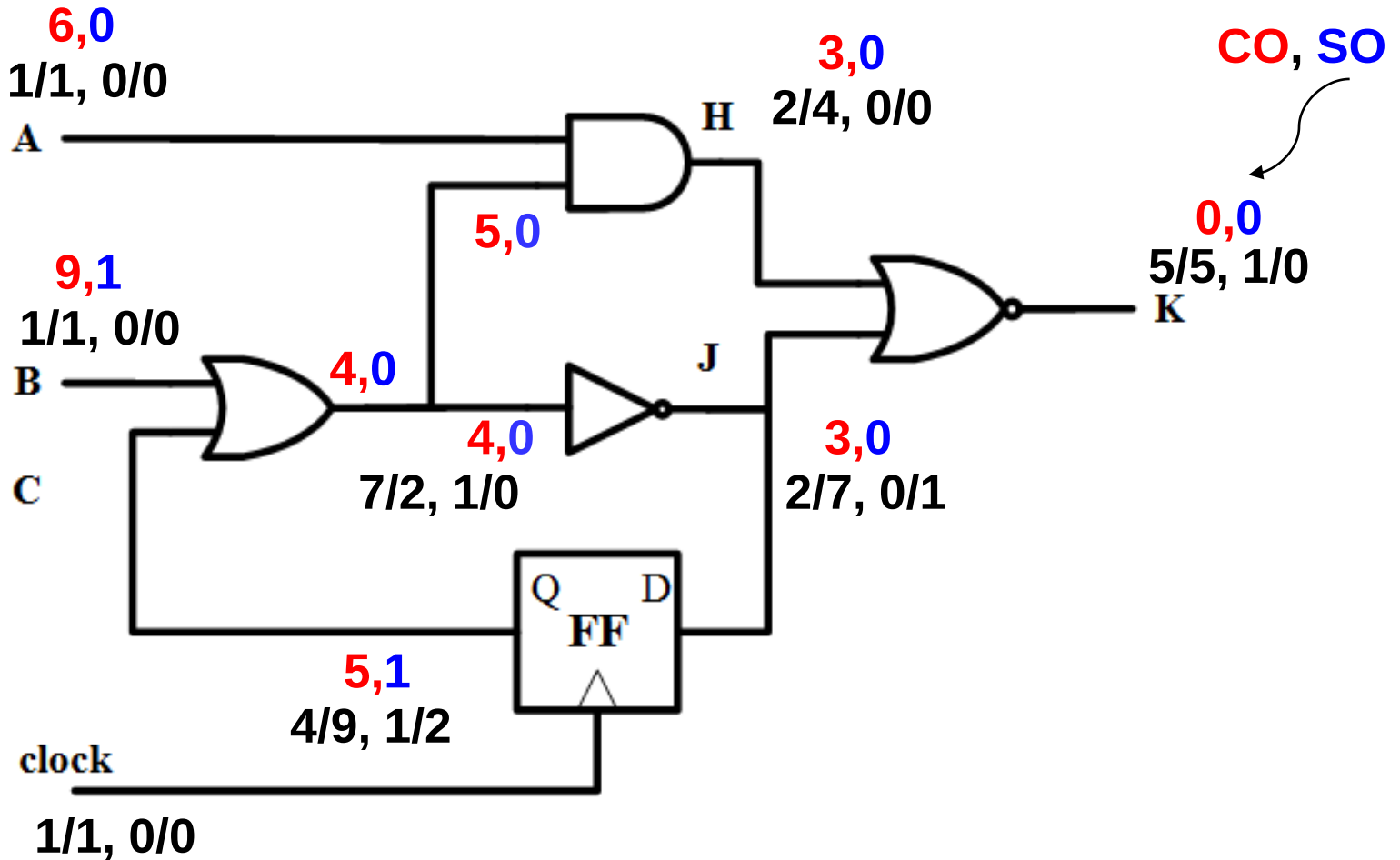
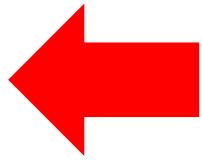


What Does $SC^1=2$ Mean?

- Control C to zero is easier. Assign $B=1$ and pulse **one** clock
- Control C to one is more difficult. Assign $B=1$ and $B=0$. **Two** clocks

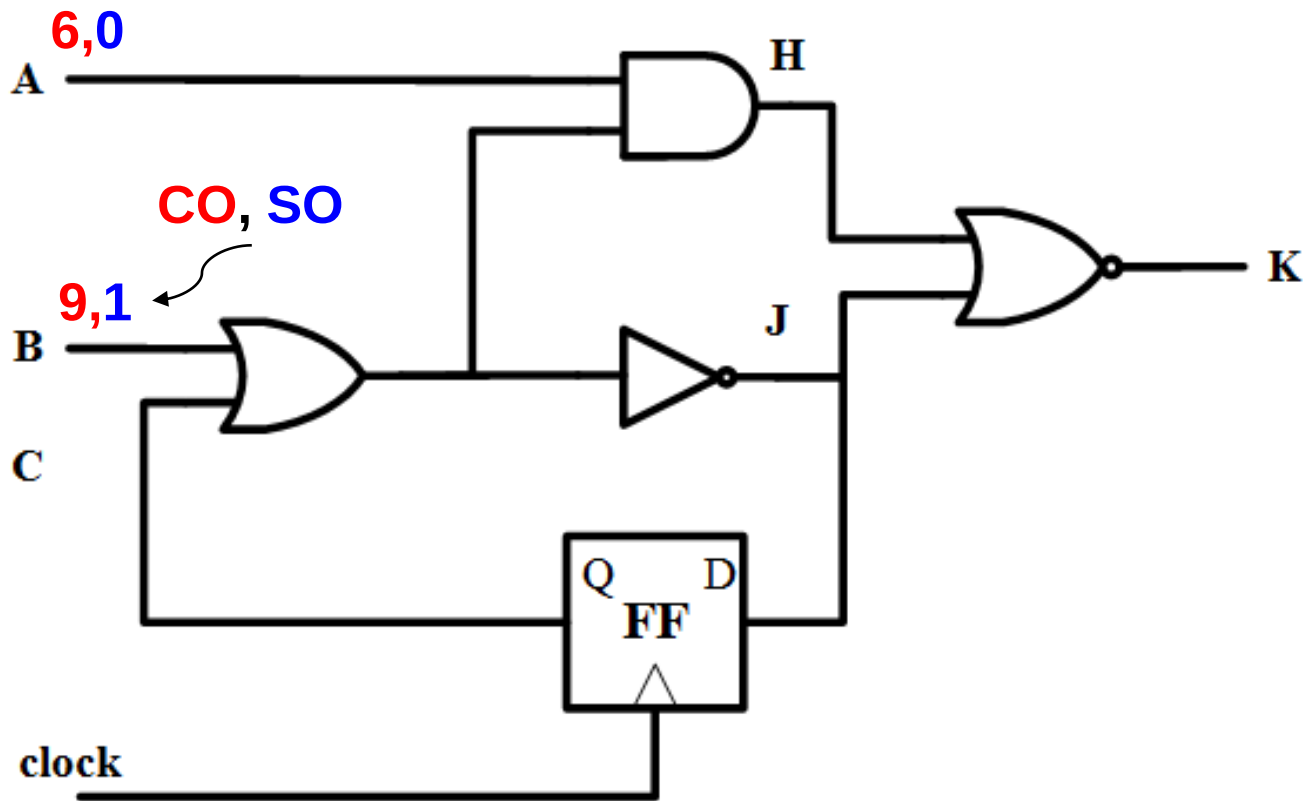


Observability Computation



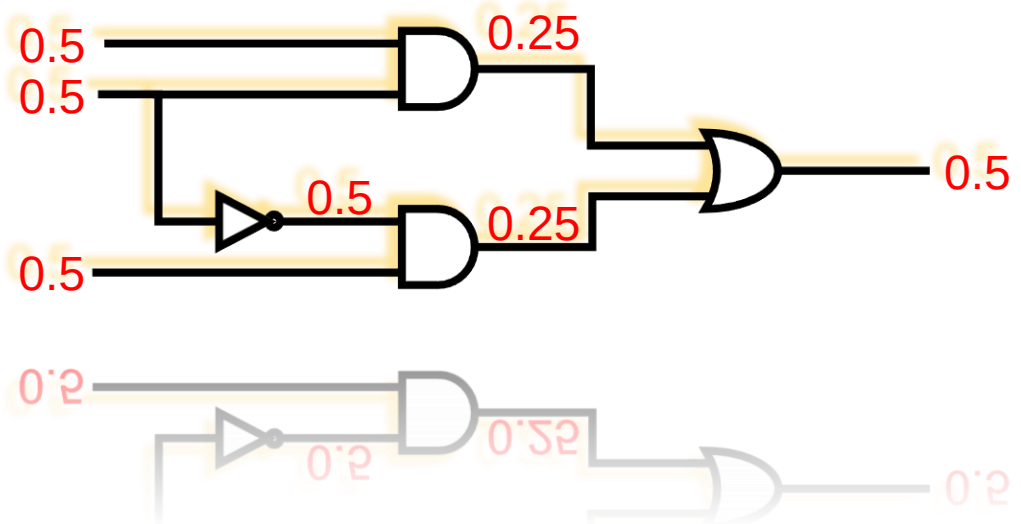
Quiz

Q1: Generate a sequence of test patterns to observe A?
Q2: Generate a sequence of test patterns to observe B?
(assume no scan. can only assign PI)



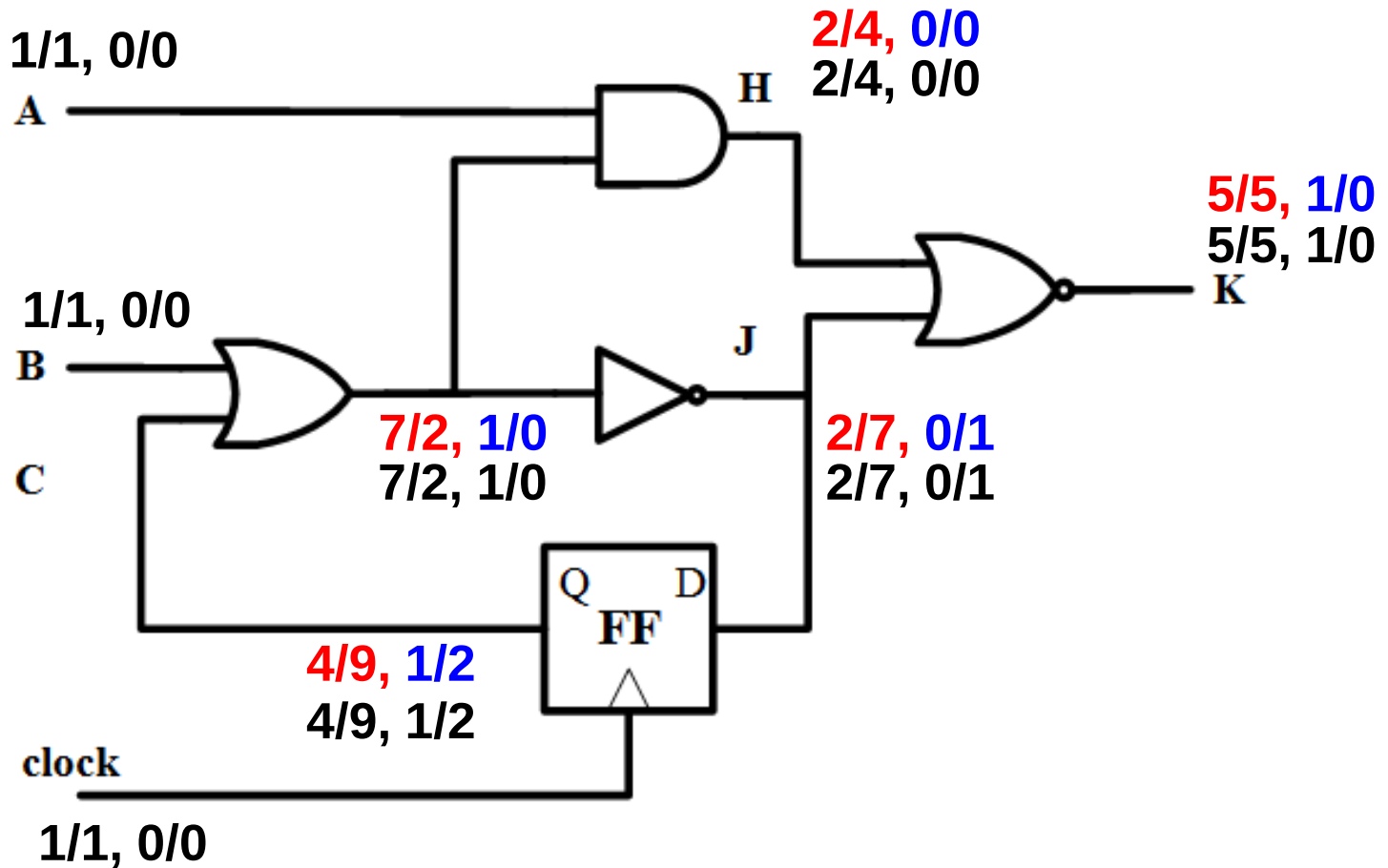
Summary

- Sequential SCOAP
 - ♦ SC^0 $SC^1 = 0$ and 1 controllability; SO = observability
 - ♦ Measure **FF assignments** (or **clock cycles**) needed
 - ♦ **Smaller** means **easier**
 - ♦ No scan is allowed
- Requires **iterations** to compute SC



FFT

- When does the algorithm fail to converge?



Computing Sequential SCOAP

- Computation of $SC^0(N)$, $SC^1(N)$, and $SO(N)$ is similar to
 - ♦ $CC^0(N)$, $CC^1(N)$, and $CO(N)$.
- Differences are
 - ① Increments sequential SCOAP by 1 only when signals propagate from **FF** inputs to Q, or backwards
 - ② May require *iterations* for controllability to converge